BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 090 · Lesson 090 · Inside City Hall: ALU, Control Unit and Registers

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does CU stand for?

Q2. What is the ALU?

Q3. What does ACC typically store?

Q4. A full adder has A=1,B=1 and carry-in=0. What are Sum and carry-out?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Control Unit — Directs instruction execution.

ALU — Performs arithmetic and logic.

ACC — Intermediate calculation register.

PC — Address of next instruction.

New learning

Observe — Control Unit generates control signals and interprets the fetched instruction. It is not the component that performs arithmetic itself.

Explain — ALU completes arithmetic and logic such as addition and comparisons; the ACC may hold a result in an accumulator-style model.

Evaluate — Registers serve distinct roles: PC next instruction, MAR memory address, MDR transferred contents, CIR current instruction, ACC intermediate calculation.

Worked example

ACC starts5; toy instruction ADD 7 is executed. Trace ALU and ACC.

1. Model response — ACC holds intermediate result, not memory address.

We Do · Practise with guidance

ACC starts 14; toy instruction SUB 6. What changes and why?

Model response

Do not say PC is overwritten by subtraction; it tracks instruction address.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

For ACC=3 and toy ADD9, state ALU calculation, ACC result, CU role. [6]

Question 2

Explain different roles of PC, MAR and CIR in a memory fetch. [6]

Question 3

Pupil claims CU performs calculations while ALU stores next instruction address. Correct both errors. [6]

Exit Ticket · Answer before checking

The Program Counter is the ALU register storing calculation results.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Control Unit.

Q2. Arithmetic Logic Unit.

Q3. Intermediate arithmetic/logic result.

Q4. $1+1+0=10_2$, so Sum0 and carry-out1.

You Do pages · Independent questions

Q1. ALU 3+9 [2], ACC12 [2], CU coordinates/decodes [2].

Q2. PC next instruction address [2], MAR selected memory address [2], CIR holds fetched instruction [2].

Q3. ALU calculates [2], CU controls/decodes [2], PC stores next instruction address [2].

Exit Ticket · Model answer

False. The ACC may store results; PC stores next instruction address.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: